Use Case Diagram — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Project: Online Examination System with Multiple-Choice Questions and Evaluation
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The actors, use cases and «include» / «extend» relationships of the Online Examination Portal. Each use case realises the functional requirements listed against it in `docs/SRS.pdf`; its step-by-step flow is in `docs/Use_Case_Flow.pdf`, and its test cases are in `docs/Test_Plan.pdf`. The SRS, the use case model and the Test Plan share one set of identifiers.

1. Notation

Symbol	Meaning
Stick figure	**Actor** — a role outside the system boundary that interacts with it
Ellipse	**Use case** — a complete, observable service the system provides to an actor
Enclosing box	**System boundary** — everything inside is built by this team
Solid line	**Association** — this actor takes part in this use case
Dashed arrow, «include»	The base use case **always** performs the included one; the arrow points at the included use case
Dashed arrow, «extend»	The extending use case runs **only under a stated condition**, at a named extension point of the base use case; the arrow points at the base use case

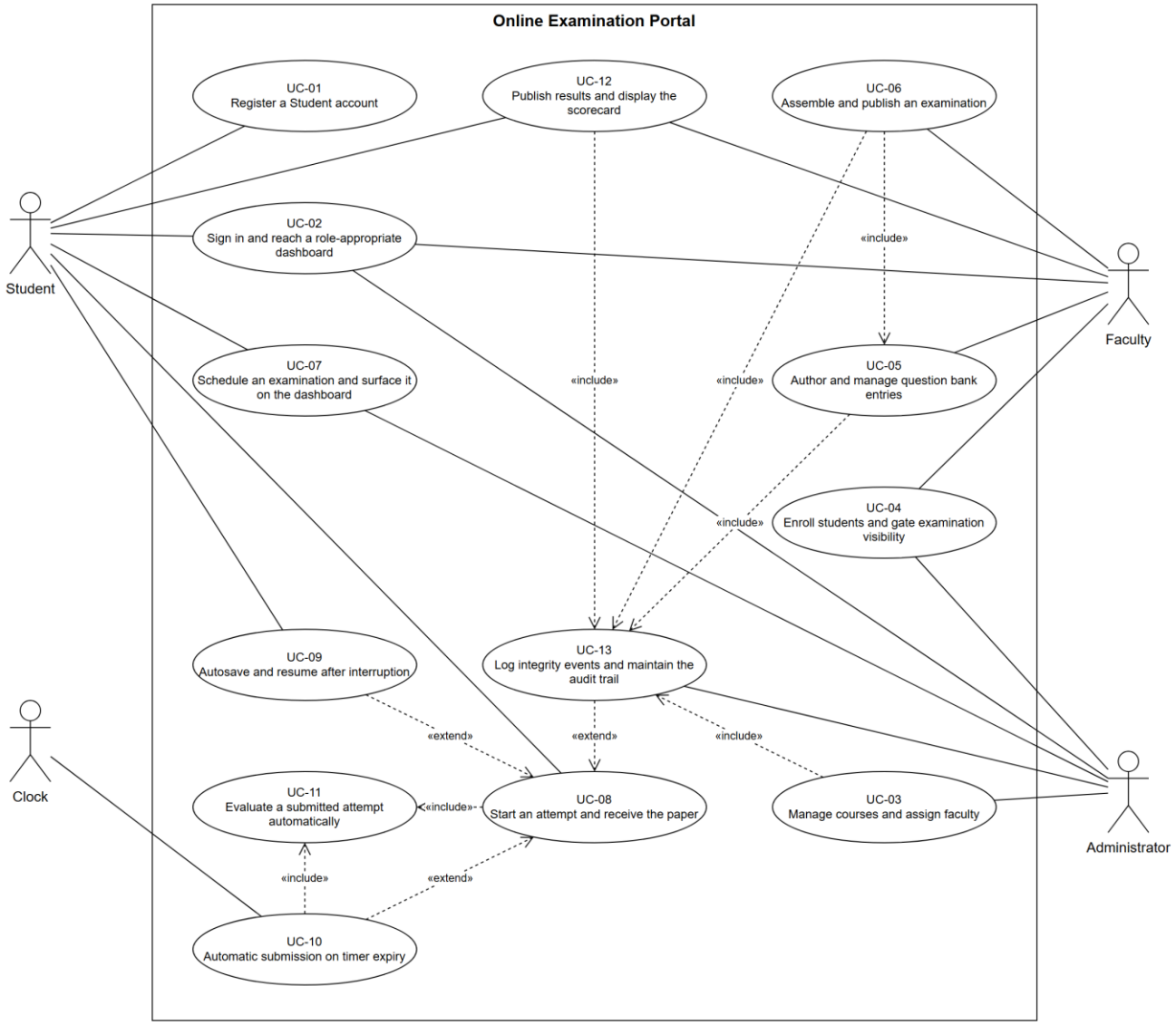
2. Actors

Actor	Kind	Description	Requirements
Student	Primary, human	An enrolled candidate. Registers, signs in, sits examinations and views published results.	FR-001, FR-006, FR-014, FR-015 … FR-021, FR-027
Faculty	Primary, human	Authors questions and examinations, sends them to the Administrator for scheduling, enrolls students, publishes results.	FR-005, FR-007 … FR-012, FR-026
Administrator	Primary, human	Maintains courses, assigns Faculty to courses, enrolls students, schedules (and so publishes) the examinations Faculty send or returns them with a note (CR-02), reviews the audit trail.	FR-004, FR-005, FR-012, FR-013, FR-030
Clock	Secondary, non-human	The server clock — the only actor that triggers behaviour with no human present: it drives automatic submission when an attempt's remaining time reaches zero.	FR-017, FR-020

Why "System" is not an actor. Several requirements — automatic evaluation (FR-022 … FR-024), withholding results (FR-025), integrity logging and audit (FR-028 … FR-030) — have no human trigger. An actor in UML is by definition *outside* the system, so the system cannot be an actor of itself. These requirements appear instead as:

- **automatic evaluation** — UC-11, reached by «include» from the submission that triggers it, so it needs no human trigger and can never be skipped;
- **withholding results** — a business rule constraining UC-12, not a use case (see §5);
- **integrity logging and audit** — UC-13, reached by «extend» from the attempt and by «include» from every use case that changes stored data;
- **the countdown and its expiry** — driven by the **Clock**, which genuinely is external to the application.

3. Use case diagram



ID	Use case	Primary actor	Requirements	Test cases
UC-01	Register a Student account	Student	FR-001	UT_001–UT_006, IT_001–IT_002, ST_001
UC-02	Sign in and reach a role-appropriate dashboard	Student, Faculty, Administrator	FR-002, FR-003	UT_007–UT_011, IT_003–IT_006, ST_002
UC-03	Manage courses and assign faculty	Administrator	FR-004	UT_012–UT_016, IT_007–IT_008, ST_003
UC-04	Enroll students and gate examination visibility	Faculty, Administrator	FR-005, FR-006	UT_017–UT_021, IT_009–IT_011, ST_004
UC-05	Author and manage question bank entries	Faculty	FR-007, FR-008, FR-009	UT_022–UT_027, IT_012–IT_014, ST_005
UC-06	Assemble and publish an examination	Faculty	FR-010, FR-011, FR-012	UT_028–UT_033, IT_015–IT_016, ST_006
UC-07	Schedule an examination and surface it on the dashboard	Administrator (CR-02), Student	FR-013, FR-014	UT_034–UT_038, IT_017–IT_018, ST_007
UC-08	Start an attempt and receive the paper	Student	FR-015, FR-016, FR-017, FR-018	UT_039–UT_044, IT_019–IT_021, ST_008
UC-09	Autosave and resume after interruption	Student	FR-019, FR-021	UT_045–UT_048, IT_022–IT_025, ST_009
UC-10	Automatic submission on timer expiry	Clock	FR-020	UT_049–UT_052, IT_026–IT_028, ST_010
UC-11	Evaluate a submitted attempt automatically	— (included, no actor)	FR-022, FR-023, FR-024	UT_053–UT_058, IT_029–IT_031, ST_011
UC-12	Publish results and display the	Faculty, Student	FR-025, FR-026, FR-	UT_059–UT_063, IT_032–

ID	Use case	Primary actor	Requirements	Test cases
	scorecard		027	IT_034, ST_012
UC-13	Log integrity events and maintain the audit trail	Administrator	FR-028, FR-029, FR-030	UT_064–UT_068, IT_035–IT_037, ST_013

Coverage: 30 of 30 functional requirements. Every requirement in the SRS appears against exactly one use case, so the model is complete and no requirement has two owners.

4. Relationships

An «include» or «extend» is drawn only if it is true of the built system; each is justified by a requirement.

From	Type	To	Condition / justification
UC-06	«include»	UC-05	FR-011 — a paper is assembled **by selecting** bank questions; there is no way to author a question inside an examination, which keeps the bank reusable across papers
UC-08	«include»	UC-11	FR-022 — every submitted attempt is evaluated automatically, with no human intervention
UC-10	«include»	UC-11	An auto-submitted attempt is evaluated on exactly the same path as a manual one (`IT_028`)
UC-09	«extend»	UC-08	Extension point *session interrupted* — FR-021; resume applies only after an interruption, so a clean attempt never takes this branch
UC-10	«extend»	UC-08	Extension point *remaining time reaches zero* — FR-020; does not occur if the Student submits first
UC-13	«extend»	UC-08	Extension point *examination window loses focus* — FR-028; nothing is recorded if focus is never lost (`UT_066`)
UC-03, UC-05, UC-06, UC-12	«include»	UC-13	FR-030 — every creation, modification and deletion of a user, question, examination, answer key or result writes an append-only audit entry
every use case except UC-01	«include»	UC-02	FR-003 — every screen and endpoint is restricted to authorised roles. Stated here as a rule rather than drawn as eleven identical arrows

The difference between the two kinds matters when the system is built: an «extend» is a branch that may never be taken in a clean attempt, whereas the «include» of UC-11 must run on **every** submission, manual or automatic.

5. Boundary notes

- **No actor is associated with UC-11.** That is deliberate: evaluation has no human trigger and no human approval step (FR-022). It is reachable only through the «include» from a submission.
- **FR-025 is a constraint, not a use case.** Withholding an evaluated result until Faculty publishes it is a rule governing when UC-12 discloses data. It has no actor-visible steps of its own, so modelling it as a use case would invent a service that nobody invokes.
- **Only a Student self-registers.** Faculty and Administrator accounts are provisioned, never self-served — FR-001 names the Student alone, and FR-003 gives every account exactly one role.
- **The Administrator is not a super-user.** The Administrator maintains courses, enrolls students schedules the papers Faculty send (CR-02) and reads the audit trail, but does **not** author questions, publish results or sit examinations. The absence of those associations is the model asserting that, not an omission.
- **There is no Proctor actor.** Integrity monitoring is automated and evidentiary: the system logs and flags; a human decides.
- **Out of scope, therefore absent:** descriptive answers and manual grading, webcam or AI proctoring, biometric identity verification, payments, adaptive testing, native mobile applications, offline mode, plagiarism detection.